

University Of Engineering and Technology, Lahore
Computer Engineering Department

Course Name: pf	Course Code: CMPE-112L
Assignment Type: lab	Dated: 07-06-2026
Semester: 1	Session: Spring 2026
Lab/Project/Assignment #: 15	CLOs to be covered: CLO 3
Lab Title: GUI using Qt designer	Teacher Name: Hafiz Muhammad Mubashar

Lab Evaluation:

CLO1	Apply appropriate programming techniques to create executable programs to solve well defined problems					
	Level1	Level2	Level3	Level4	Level5	Level6
(5)						
Total						/5

Rubrics for Current Lab Evaluation

Scale	Marks	Level	Rubric
Excellent	5	L1	Submitted all lab tasks, BONUS task, have good understanding.
Very Good	4	L2	Submitted the lab tasks but have good understanding
Good	3	L3	Submitted the lab tasks but have weak understanding.
Basic	2	L4	Submitted the lab tasks but have no understanding.
Barely Acceptable	1	L5	Submitted only one lab task.
Not Acceptable	0	L6	Did not attempt

LAB No 1

Lab Goals/Objectives:

- To understand the concept of Graphical User Interface (GUI) development using Qt Designer.
- To learn how to design user-friendly interfaces using Qt Designer tools and widgets.
- To understand how to connect GUI components with Python code.
- To develop desktop applications using the PyQt framework and Qt Designer.

Equipment Required:

- Laptop/PC
- Python 3.x
- Qt Designer
- PyQt5 / PyQt6
- VS Code (or any Python IDE)
- Windows Operating System

Theory:

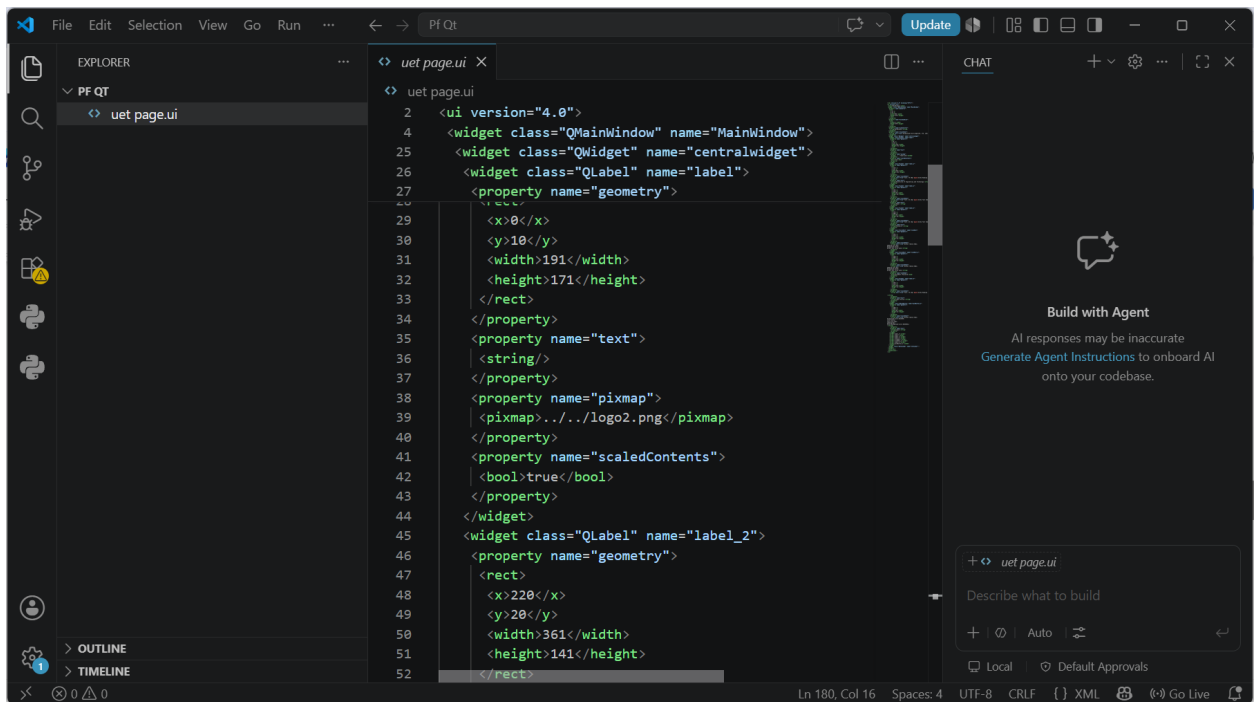
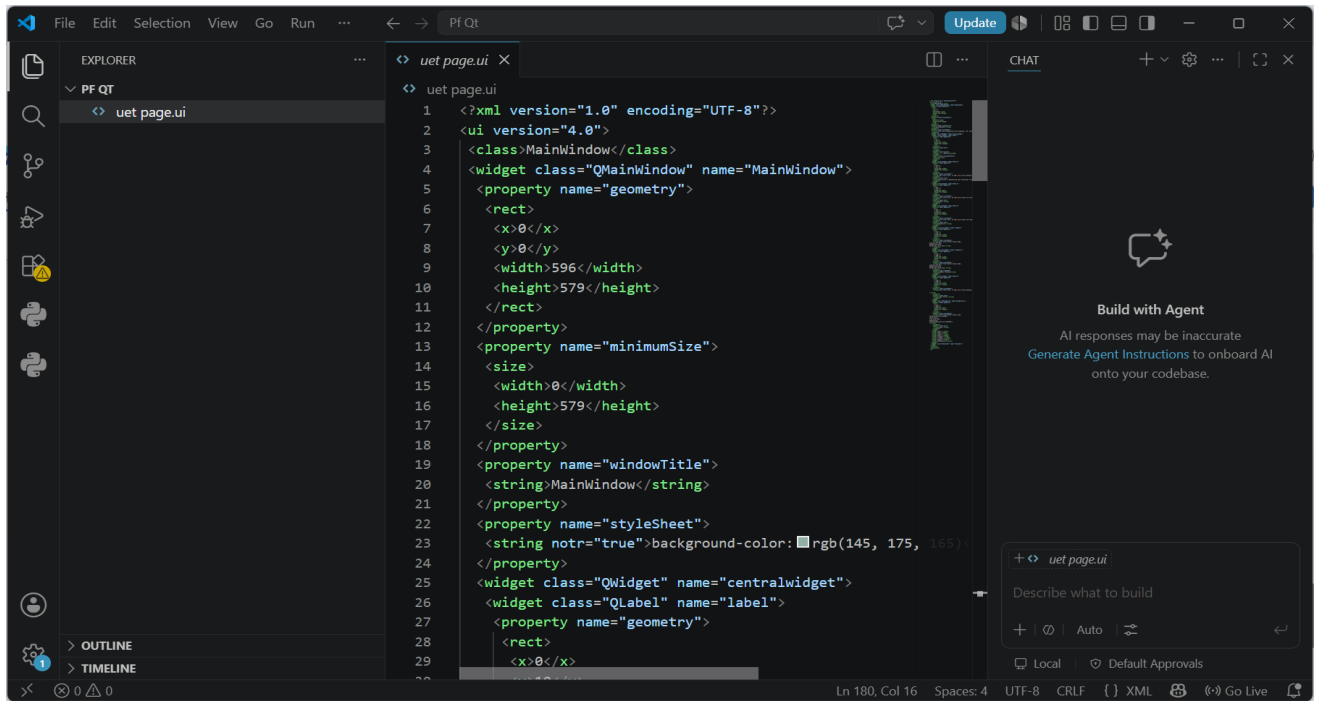
A **Graphical User Interface (GUI)** allows users to interact with an application through visual components such as buttons, labels, text boxes, menus, and windows instead of command-line instructions. **Qt Designer** is a drag-and-drop tool used to design GUI layouts quickly and efficiently without writing interface code manually. The designed interface is saved as a **.ui** file, which can be loaded into Python applications using the **PyQt** library. Qt Designer simplifies the development of desktop applications by providing a visual environment for creating attractive and user-friendly interfaces.

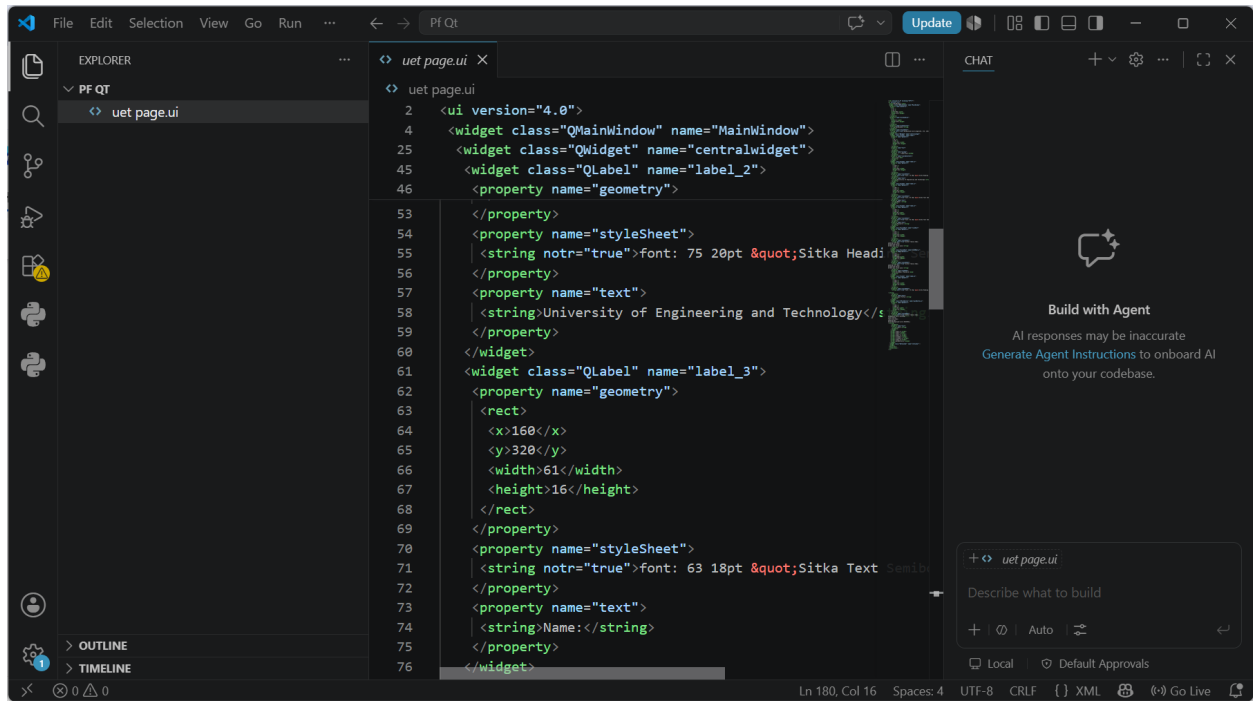
Lab Work:

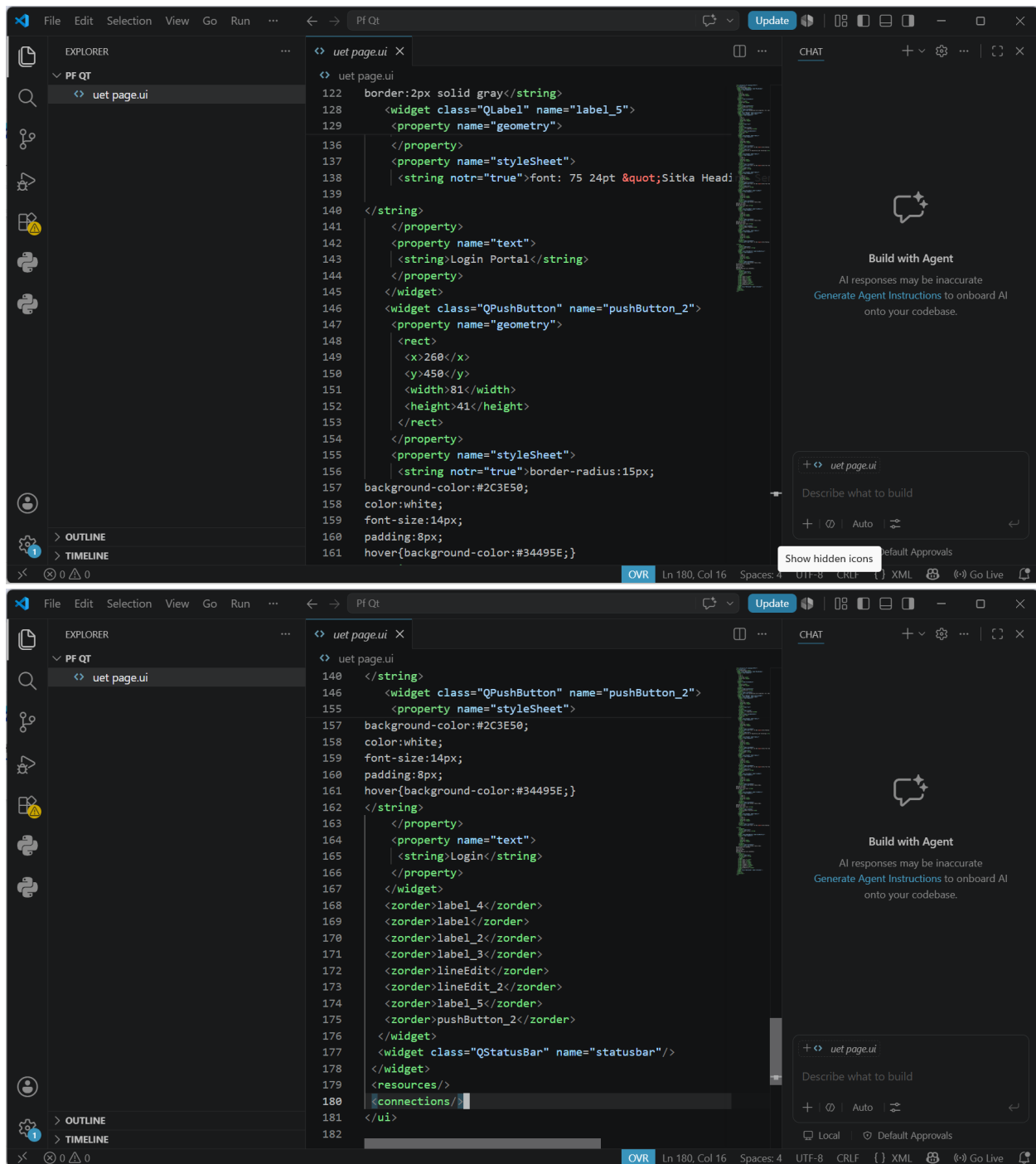
Task 1:

Write a program to design a Login Portal using Qt Designer by adding labels, line edit fields, a login button, and an image. Apply style sheets to enhance the user interface and configure the password field to hide user input.

Result:







Conclusion:

In this lab, we learned how to design a GUI application using **Qt Designer**. We created a login interface by placing and configuring different widgets, applied style sheets to improve the user interface, displayed an image, and secured the password field using the password echo mode. This lab enhanced our understanding of GUI design and the development of user-friendly desktop applications using **PyQt** and **Qt Designer**.